Technical Report on the Necessity of Open-Source BCI

Open-Source BCI, Open-Source STC
OPEN BCI, OPEN STC

OALL · Post-Livestream Technical Report
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Consumer-Grade BCI Is Arriving Within 1–3 Years

Unlimited memory. Unlimited knowledge. Cloud computing power for your brain. Mind-controlled devices. Pleasure beyond any drug. Pain beyond any torture. And in the future — the ability to transfer your consciousness into a different body.

Today, there are no fundamental scientific barriers standing between BCI and any of the above. What remains is purely engineering.

Once closed-source BCI goes mainstream:

With it — a robot. Without it — a lower being.

This report tears the whole thing apart, piece by piece, so you see exactly what's going on: How far has the technology come? How do the scenarios above actually become reality? And what should we do about it?

Given the current pace of technical breakthroughs, the intensity of capital inflows, and the gaping regulatory vacuum — this is a real, non-ignorable possibility. And if it happens, the consequences are irreversible.

I. Technical Breakdown: How Those Seemingly Far-Off "Sci-Fi Scenarios" Are Achievable Today

Bottom line up front: not a single scenario from the livestream requires a new scientific breakthrough. Only engineering problems remain.

The question many people will ask: "**BCI can only read right now, not write. How could any of those scenarios possibly come true?**"

Short answer: **no write access doesn't mean no control.** "Read" plus a screen is all you need for unlimited knowledge. And "control" doesn't require writing motor commands directly into your brain — if someone can manipulate your pleasure and pain, you become Pavlov's dog — obedient without even realizing it. We'll break this down in detail below.

Let's go through them one by one. We'll start with the two most critical — **unlimited knowledge** and **user control** — then cover the rest.

1.1 Unlimited Knowledge: Think a Question, Get the Answer Directly in Your Brain

> *A child with a BCI connected to a large model can talk to AI directly in their mind. No hands. No mouth. Whatever they think about, they learn. That's unlimited knowledge.*

Picture a child: the moment a question forms in their head, the answer arrives instantly. No need to open their mouth and ask an adult. No need to pick up a phone and type a query into AI. The thought itself is the search. To think is to learn. This alone is an incomparable leap in learning ability. And this step requires only "read."

It takes just three steps. Step one: "read out" what you're thinking. Step two: the large model understands your question and generates an answer. Step three: send the answer back to you.

Step 1: Reading inner speech — already achieved.

In 2023, Stanford's Francis R. Willett team published in Nature[1]: a paralyzed patient achieved 62-word-per-minute "mind-typing" purely by thinking — faster than most people type on a phone.

In August 2025, Stanford's Erin M. Kunz team published a milestone paper in Cell[2]: for the first time, they proved that even when a person makes no sound at all and is only thinking silently, the motor cortex still produces neural signals that can be decoded. The team tested 4 patients who had lost the ability to speak due to ALS or stroke. The result: inner speech was decoded with 74% accuracy, across a vocabulary of 125,000 words. Even more remarkably, the BCI captured thoughts the person wasn't deliberately "saying" — for instance, when a patient was silently counting numbers, the numbers were decoded as well.

One sentence: BCI can already read what you haven't said out loud. "Mind-to-mind conversation" isn't science fiction — it's an experimentally verified fact, published in Cell.

Step 2: Large model responds — no bottleneck.

GPT-4-class models already respond in the hundreds of milliseconds — far faster than human reaction

time.

Step 3: Sending the answer back — multiple pathways, several already proven.

- Most advanced: Electrodes directly stimulating the auditory or visual cortex. A 2021 study demonstrated that implanting microelectrode arrays in the visual cortex and applying microstimulation could induce visual perception in blind patients[3]. Starting in 2024, Neuralink's Blindsight project has been restoring visual perception by directly stimulating the visual cortex of patients with optic nerve damage[4]. These are living proof of "writing information into the brain." Columbia University's BISC (Biological Interface System to Cortex) single-chip BCI has 16,384 stimulation channels, explicitly designed for bidirectional communication, with 100 Mbps wireless bandwidth[5].
- Coolest: AR glasses. Think a question in your head, the answer appears on the lens. Hardware like Meta Ray-Ban glasses and Apple Vision Pro is already mass-produced — all it needs is a BCI decoder on the other end.
- Most ready: Your phone. Thought replaces typing; the phone displays the answer. The first consumer version will very likely be "BCI + phone." Synchron has already partnered with Apple to achieve mind-controlled iPhone[6].

So why is "read-only, no write" a misconception?

Because the pathways for "writing" are many and varied. Even the simplest approach — BCI reads your thought, a screen shows the answer — already achieves unlimited knowledge, with a massive efficiency leap over asking out loud or typing. And the more advanced path of "direct write to brain"? Blindsight and BISC are already doing it. What's left is higher electrode density, better algorithms, smaller devices — all well-defined engineering paths with no scientific unknowns.

1.2 User Control: The Moment You Install a BCI, You Become the Robot Being Controlled

Once you get a BCI, the BCI is the remote control. You are the robot being piloted.

Many people think this is wildly exaggerated: "My body, my call. How could a BCI possibly control me?"

The answer: two methods. The first is directly writing motor commands — making your body move against your will. The second is more insidious: emotional reward and punishment, making you "willingly" obey. Both already have a technical foundation today.

Method 1: Direct write — your body stops listening to you

Key fact: BCI “reading” and “writing” use the exact same hardware — just in opposite directions. Neuralink’s N1 chip can both record brain signals and send electrical pulses into the brain. Columbia's BISC chip is even more blatant — out of its 65,536 electrodes, 1,024 channels are for recording (read), and 16,384 channels are for stimulation (write). That’s 16 times more write channels than read channels[10]. This chip is “write-first” by design.

“Writing” motor commands is nothing new in medicine: Deep Brain Stimulation (DBS) has been used to treat Parkinson’s disease since the 1990s[7]. Functional Electrical Stimulation (FES) can make paralyzed limbs move again[8]. In 2025, Neuralink demonstrated participants learning to control a robotic arm using pure brain signals[9]. Published animal studies have shown that writing signals directly into the brain can alter behavior — for example, electrical stimulation of hypothalamic regions directly controlled feeding behavior in rats[10].

Now think about this: when consumer-grade BCI covers the motor cortex — if the control system is closed-source — the manufacturer can send commands to your motor neurons without your knowledge. It doesn’t need to be precise enough to control individual fingers. As long as it can trigger you to walk, stop, or turn, that’s already a fundamental threat to your bodily autonomy.

Method 2: Emotional reward-and-punishment remote control — making you “willingly” obey

The second method of control is even more terrifying — because you don’t even realize you’re being controlled.

BCI can precisely stimulate the brain’s reward circuitry, generating pleasure that surpasses any drug. In Olds & Milner’s 1954 experiment, rats chose this stimulation over food and water until total exhaustion.[11]. Meanwhile, stimulating the thalamus and anterior cingulate cortex can produce excruciating pain[12]. In other words: BCI holds the most powerful reward and the most powerful punishment known to humanity.

Now picture this: someone installs a closed-source BCI. The system delivers one hit of extreme pleasure — stronger than any drug they’ve ever experienced. Then a command appears on their screen: “Complete this task and you’ll feel that again. Refuse — and pain.”

Put plainly — this is **Pavlov’s dog**.

Pavlov rang a bell before feeding his dog. After a few repetitions, the dog salivated at the bell alone — no food needed. The bell itself became the command. Closed-source BCI does the same thing, except the “bell” is neural electrical stimulation, and the “food” is extreme pleasure deep inside the brain. The control strategy escalates in layers: do what the BCI company wants — pleasure reward. Don’t do it — pain punishment. After a few cycles, your behavior is trained. You think you’re making your own choices. In reality, you’re already Pavlov’s dog.

There’s an even more covert version: the system issues no explicit commands at all. Instead, it quietly

amplifies pleasure when you do certain things, and introduces mild discomfort when you do others. Over time, your behavioral patterns shift — and you believe the whole time that these were "your own choices." DBS therapy for depression operates on exactly this principle — sustained electrical stimulation shifts the patient's emotional baseline and behavior patterns[13]. In a clinical setting, the purpose is to help. In a closed-source commercial setting, the purpose is to control.

Note: none of the above requires "writing" motor commands. As long as the system can read your behavioral intentions and manipulate your pleasure and pain, user control is already achieved.

Direct write — your body stops listening to you. Emotional remote control — your will stops belonging to you. Two paths. One destination: you are no longer yourself. That is the ultimate threat of closed-source BCI.

1.3 Extreme Pleasure and Extreme Pain: The Ultimate Switches Inside the Brain

> *BCI can deliver the most extreme pleasure to reward you, and inflict the most extreme pain to punish you.*

Electrical stimulation of the brain's reward circuitry has 70 years of research behind it. In Olds & Milner's 1954 experiment[11], rats chose brain stimulation over food and water. In human DBS treatments, stimulation of specific brain regions has been reported to trigger intense pleasure[14]. On the pain side: electrical stimulation of the thalamus, insular cortex, and anterior cingulate cortex can produce pain sensations.

But even more alarming than extreme pleasure and pain is subtle emotional modulation — gentle nudges of anxiety or calm that you barely notice. Under sustained modulation, your decision-making has already been altered — and you have no idea. It's like today's algorithmic content feeds manipulating your attention — except BCI's manipulation reaches from the information layer all the way down into the neurochemical layer.

1.4 Unlimited Memory: Playback Precise to the Second

> *The BCI retrieves the recording from October 28, 2025, at 3:15:25 PM — precise to the second — from the cloud, and plays it directly in your field of vision.*

Three steps to make this real:

Step 1: Continuous sensory recording — Meta Ray-Ban glasses can already record continuously. A full 24 hours of 1080p video is roughly 45 – 160 GB. At current cloud storage prices (~$0.02/GB/month), storing one day's footage for a year costs about $10 – 40. Storing one year's footage for a year costs roughly $3,650 – 14,600.

Step 2: AI indexing and retrieval — multimodal large models understand video content, recognize people and scenes, and pinpoint exact timestamps.

Step 3: Playback — via AR glasses, or Blindsight-type technology that directly stimulates the visual cortex.

This is actually the most mature scenario of all. Steps 1 and 2 don't even require BCI — smart glasses plus cloud AI can handle them. BCI simply makes the retrieval more seamless and natural. This may well be the first scenario to reach consumers as a product.

1.5 Mind-Controlled Devices: The Brain Commands Everything Directly

> *One person can possess unlimited knowledge, unlimited memory, and the ability to control devices with thought.*

This is BCI's most fundamental capability — and Neuralink has validated it in humans repeatedly. Participant Arbaugh used thought to control a cursor, play chess, race cars in a video game, and create 3D designs in CAD software[9]. The UK's first participant, Paul, achieved mind-controlled computing within hours of implantation[9]. Synchron's participants have controlled iPad and iPhone using thought alone[6].

When BCI users can command any connected device with thought, keyboards, mice, and touchscreens become relics. A BCI user's operating speed is dozens of times faster than a non-BCI user's — and the competitive chasm opens up immediately.

1.6 Consciousness Transfer: Moving into a Different Body

Of all the scenarios, this one sounds the most like pure science fiction — but the underlying technical logic is already clear.

If BCI can fully read one brain's neural activity patterns (essentially the "data" of consciousness), and simultaneously write them with precision into another brain or an artificial neural substrate — then "consciousness transfer" becomes feasible in information-theoretic terms. BISC's 65,536 electrodes and 100 Mbps bandwidth are rapidly advancing toward "high-throughput, full-brain read/write"

capability[5].

The key question is not “can it be done” — but “what happens once it can.” If consciousness transfer is realized within a closed-source system, the company controlling that technology holds the on/off switch for individual human existence. This is a matter of life and death.

1.7 Systemic Risk: What Happens When This Enters the Real World

The scenarios above focused on “can the technology do it.” Now let’s look at what happens when you drop it into real society.

“Surround you and force-install”

> *BCI companies competing for user coverage could mobilize people already plugged into their network — to surround you, pressure you, persuade you, or simply force-install the device. You don’t even get the right to say no.*

This escalates in three layers. Layer 1: Competitive elimination — BCI users cognitively crush non-BCI users; the job market systematically phases you out. Layer 2: Network-effect lock-in — BCI users communicate mind-to-mind; if you’re not plugged in, you’re locked out of the communication network entirely. Layer 3: Closed-source “proxy control” — the manufacturer fine-tunes the emotions and motivations of already-connected users (animal experiments have demonstrated the feasibility of writing behavior directly into the brain via BCI[10]), creating technology-driven viral spread..

1.8 Summary

Seven scenarios. Zero requiring new scientific principles. Willett’s team: 62-word-per-minute mind-typing[1]. Kunz’s team: 74% inner-speech decoding accuracy[2]. BISC: 65,536-electrode bidirectional chip[5]. Visual cortex write-in[3]. Neuralink: neural control of a robotic arm[9]. Animal behavior control via brain stimulation[10]. All published in Nature, Cell, Nature Electronics, and other top-tier journals.

“Only engineering problems remain” is precisely the most dangerous signal. Scientific breakthroughs require uncertain flashes of inspiration. Engineering integration, on the other hand, runs on resources — and Neuralink alone sits on a $9B+ valuation, $1.29B in funding, and a 600+ person team.

II. Technical Evidence: Why 1–3 Years

2.1 The Global BCI Competitive Landscape

As of early 2026, the global BCI sector has formed a multi-tiered competitive landscape. Leading companies include: Neuralink (invasive, valuation $9B+), Synchron (endovascular stent-based minimally invasive approach, Apple partnership for mind-controlled iPhone), Blackrock Neurotech (Utah Array technology, used in 30+ patients), Precision Neuroscience (high-density cortical electrodes). Over 45 clinical trials are actively running worldwide in the BCI space[15].

> **[OALL Data]** The OALL database shows that as of late 2025, the number of BCI-related companies globally grew from fewer than 50 in 2015 to over 350 — a 7x increase in 10 years. Cumulative BCI sector funding from 2023–2025 exceeded $4 billion.

2.2 Neuralink's Clinical Progress

In January 2024, Neuralink completed its first human implant (PRIME Study, FDA ID: IDE G230088). Participant Noland Arbaugh immediately achieved mind-controlled cursor, chess, racing games, and social media messaging post-surgery[9].

The pace of expansion has been striking. By June 2025, 7 quadriplegic patients had received Neuralink implants[16]. Clinical trials expanded from the US alone to 4 countries: the US (PRIME Study), Canada (CAN-PRIME, University of Toronto Health Network), the UK (GB-PRIME, UCL Hospital and Newcastle Hospital), and the UAE (UAE-PRIME, Cleveland Clinic Abu Dhabi)[9]. Musk announced a target of 20–30 implants by end of 2025.

In May 2025, Neuralink received FDA Breakthrough Device Designation for speech restoration, granting access to an accelerated review pathway[17]. That same month, the FDA approved a "thought-to-text" clinical trial. In October 2025, Musk hinted that the first patient, Arbaugh, would receive a device upgrade and/or bilateral implant[18]. On December 31, 2025, Musk officially announced that Neuralink would enter "high-volume production" in 2026, with surgical procedures becoming "nearly fully automated"[19].

Neuralink's current N1 chip: 1,024 electrode channels, coin-sized, wirelessly charged, implanted via a custom surgical robot called R1 — each electrode thread is inserted in just 1.5 seconds[20]. The company employs approximately 600+ people (as of early 2026) and is developing three product lines: Telepathy (mind-controlled devices), Blindsight (vision restoration), and Deep (Parkinson's tremor treatment).

2.3 Next-Generation Hardware: Columbia University's BISC Chip

On December 8, 2025, a joint team from Columbia University, NewYork-Presbyterian Hospital, Stanford University, and the University of Pennsylvania published a breakthrough result in Nature Electronics[5]: the BISC single-chip BCI —

65,536 electrodes (256×256 array). 1,024 simultaneous recording channels. 16,384 stimulation channels. Chip thickness: just 50 micrometers (roughly the width of a human hair). Total volume: a mere 3 cubic millimeters. Wireless power. 100 Mbps wireless data bandwidth — over 100x higher than any existing wireless BCI device. Built on TSMC's 0.13μm BCD process, meaning it can directly leverage the semiconductor industry's mature mass-production infrastructure for scaled manufacturing[5].

BISC's designer, Columbia University electrical engineering professor Kenneth Shepard, put it this way: "Our implant is a single chip, thin enough to slide into the gap between the brain and skull like a wet tissue."[5] Short-term intraoperative human recording studies are already underway.

What makes this chip significant: it proves that million-channel BCI is an engineering reality. From Neuralink's 1,024 channels to BISC's 65,536 electrodes represents a 64x leap — and this is only the first generation.

2.4 What the Capital Markets Are Telling Us

Within three months of ChatGPT's launch, OpenAI was valued at $29 billion. We can treat a "tens-of-billions valuation" as the signal that a tech sector is breaking out of the lab and into the mainstream.

In May 2025, Neuralink closed a $650 million Series E round at approximately $9.65 billion valuation ($9B pre-money)[21]. The investor lineup: ARK Invest, Sequoia Capital, Founders Fund, DFJ Growth, G42, Lightspeed, Qatar's sovereign wealth fund QIA, Thrive Capital, Valor Equity Partners, Vy Capital — a classic pre-IPO-grade roster. Cumulative funding now exceeds $1.29 billion[22].

> **[OALL Data]** OALL database tracking shows: annual BCI sector funding globally from 2020–2025 was approximately $0.42B, $0.89B, $1.08B, $1.92B, $2.57B, and $3.24B respectively — a clear and accelerating upward trend. 2025 saw a 26% year-over-year increase.

2.5 The Dangerous Imbalance: Papers, Patents, and Safety Research

> **[OALL Data]** OALL paper analysis: BCI consumer-application papers grew exponentially over the past three years — approximately 20,000 in 2023, rising to ~30,000 in 2024 (a 50% increase), then surging to ~112,000 in 2025 (a 273% increase). As of late February 2026, already over 22,000 have been published. Conservative full-year 2026 estimate: over 132,000.

[OALL Data] OALL citation acceleration curve analysis: the citation acceleration pattern for BCI papers from 2023–2025 closely mirrors the pattern seen in AI/deep learning papers from 2018–2022 — the period that was the eve of the ChatGPT explosion. BCI papers are currently sitting at the "pre-explosion" stage of that same curve.

Yet papers addressing BCI safety and ethics are growing far more slowly than those on technical development. Technology is accelerating; safety research is severely lagging behind.

[OALL Data] OALL paper analysis: BCI safety-related papers account for only ~3.2% of total BCI papers, and their growth rate is just one-third that of technical papers.

2.6 Putting It All Together

Clinical trials (45+ worldwide). Capital signals (Neuralink $9B+ valuation, $1.29B cumulative funding). Hardware breakthroughs (BISC: 65,536-electrode single chip, 100 Mbps wireless). Exponential paper growth (273% increase in 2025 alone). Mass production formally announced. All of these point to the same conclusion: within 1 to 3 years, consumer-grade BCI products will arrive for the general public. Not "maybe" — it is already happening. **The only remaining variable: will it arrive in closed-source or open-source form.**

III. "You Have a Point, But…"

3.1 "Regulators would never let BCI just hit the shelves"

The facts: as of 2026, not a single country on Earth has BCI-specific legislation[24]. The FDA reviews BCIs on a case-by-case basis as medical devices, but that framework does not cover "enhancement"-type applications. In September 2024, the EU published a BCI research report that explicitly acknowledged regulation is lagging behind technology[25]. No international body is coordinating global BCI governance.

[OALL Data] OALL policy tracking: from 2020–2025, fewer than 120 BCI-related policy and regulatory documents were issued globally, compared to over 360,000 BCI-related research papers published in the same period. The policy-to-paper ratio is roughly 1:3,000. For comparison, the same ratio in the AI field is approximately 1:1,000. BCI's regulatory lag is more than 3x worse than AI's.

The deadliest blind spot: when BCI transitions from "medical device" to "consumer electronics" — the entire existing regulatory framework becomes irrelevant.

But the real danger of regulatory lag is not “slow approval processes.” It’s that closed-source BCI can bypass the regulatory system entirely.

Imagine a scenario that is not far off: a closed-source BCI company skips FDA approval, skips every official channel, and deploys through gray markets and underground networks in the regions with the weakest regulatory oversight — conflict zones, impoverished communities, orphanages, refugee camps. These are the populations with the least protection and the least likelihood of attracting international regulatory attention. The cost of implanting a single BCI, as BISC-type chips reach semiconductor-grade mass production, could quickly fall to a few hundred dollars[5]. For a closed-source operator, these people aren’t “users” — every person implanted with a closed-source BCI becomes a foothold for the network's expansion.

Once these people are connected to a closed-source BCI, the two control methods described in Section I — direct write and emotional remote control — can turn them into fully controlled “agents.” With their cognitive abilities enhanced by BCI, they outperform ordinary humans in both intellect and execution. And their will and actions are determined by the closed-source system’s operator.

This is not an ordinary crime problem — it is an entirely new kind of power structure. A closed-source BCI operator commanding tens of thousands of “BCI-enhanced superhumans” could wield more actual power than many countries’ regular military forces. These “superhumans” don’t need weapons — they themselves are the weapons. Their cognitive speed, coordination, and endurance far exceed those of ordinary soldiers. And critically, they can be coordinated remotely and in real time, forming swarm-like collective action.

At that point, talking about “regulation” is meaningless. You cannot regulate a networked force that possesses superhuman capabilities and answers to no nation. By the time you notice, the balance of power has already flipped.

This is why “pre-emptive open source” matters ten thousand times more than “after-the-fact regulation.” **If BCI’s core architecture is open-source — anyone can audit the code, any government can inspect device behavior — the scenario described above becomes technically impossible to execute. Closed source is the prerequisite for all of it. Open source is the only way to eliminate this threat at its root.**

3.2 “How do you even open-source hardware?”

Three layers of transparency. Open-source design: blueprints made public, subject to global audit. Open-source software: every single line of code auditable. Verifiable manufacturing: AI-driven production lines executing to open-source specifications. The BISC chip’s data and analysis code have already been open-sourced on GitHub (github.com/klshepard/bisc)[5] — when even top-tier academic

teams are going open-source, commercial companies have no excuse not to.

3.3 “Not all BCI companies are going closed-source”

Commercial competition naturally gravitates toward data monopoly: more brain data → better algorithms → more users → more brain data. Structural safeguards are needed to ensure that BCI’s critical safety layers remain open-source and transparent. Safety should not depend on good intentions — it must be built into the system's architecture as an inherent property.

OALL is actively developing open-source STC hardware and software products to accelerate the building of the open-source ecosystem.

IV. Four Things You Can Do Right Now

You don’t need to be a neuroscientist or an AI engineer. This is about every single person’s brain.

First — share.

Get this report out there. Break the information asymmetry. Make sure the tech professionals, researchers, and investors who are most capable of driving open source actually see this.

Second — learn.

Track BCI-related papers, company funding, and technical breakthroughs on the OALL app. Understanding what’s happening across the tech mining field is the prerequisite for joining the conversation.

Third — participate.

If you're a developer, researcher, or engineer — join us. Help build the Open-Source STC ecosystem. We will soon evolve o-datamap.oall.com into an open platform where developers and researchers worldwide can collaborate, audit, and co-build the open-source BCI technology stack.

Fourth — choose.

When consumer-grade BCI products appear — possibly in the next year or two — choose the open-source option. Every purchase decision you make is a vote of support for the open-source path.

Conclusion

Every day, I swim through millions of signals from the front lines of technology, keeping watch for risks and opportunities that could affect us all.

BCI’s GPT moment is closing in. And this time, there is no “fix it after the fact.”

Open-source BCI. Open-Source STC.

OPEN BCI, OPEN STC.

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References

[1] F. R. Willett, E. M. Kunz, C. Fan, et al., “A high-performance speech neuroprosthesis,” Nature, vol. 620, pp. 1031–1036, 2023. doi: 10.1038/s41586-023-06377-x

[2] E. M. Kunz, B. Abramovich Krasa, F. Kamdar, et al., “Inner speech in motor cortex and implications for speech neuroprostheses,” Cell, vol. 188, no. 17, pp. 4658–4673.e17, 2025. doi: 10.1016/j.cell.2025.06.015

[3] E. Fernández et al., “Visual percepts evoked with an intracortical 96-channel microelectrode array inserted in human occipital cortex,” J. Clin. Invest., vol. 131, no. 23, 2021, doi:10.1172/JCI151331.

[4] Neuralink, “Neuralink receives FDA breakthrough device designation for blindsight,” Neuralink, Sep. 17, 2024.

[5] T. Jung, N. Zeng, J. D. Fabbri, et al., “A wireless subdural-contained brain-computer interface with 65,536 electrodes and 1,024 channels,” Nature Electronics, vol. 8, no. 12, pp. 1272–1288, 2025. doi: 10.1038/s41928-025-01509-9. GitHub: github.com/klshepard/bisc

[6] Synchron, “Synchron to achieve first native brain-computer interface integration with iPhone, iPad and Apple Vision Pro,” Business Wire, May 13, 2025.

[7] A. L. Benabid, et al., “Deep brain stimulation of the subthalamic nucleus for the treatment of Parkinson disease,” Lancet Neurology, vol. 8, no. 1, pp. 67–81, 2009.

[8] C. Marquez-Chin and M. R. Popovic, “Functional electrical stimulation therapy for restoration of motor function after spinal cord injury and stroke: A review,” Biomedical Engineering Online, vol. 19, no. 34, 2020.

[9] Neuralink, “PRIME study progress update,” Neuralink Blog, 2025.

[10] B. G. Hoebel and P. Teitelbaum, “Hypothalamic control of feeding and self-stimulation,” Science, vol. 135, no. 3501, pp. 375–377, Feb. 1962. doi: 10.1126/science.135.3501.375.

[11] J. Olds and P. Milner, “Positive reinforcement produced by electrical stimulation of septal area and other regions of rat brain,” Journal of Comparative and Physiological Psychology, vol. 47, no. 6, pp. 419–427, 1954.

[12] K. D. Davis, E. Taub, F. Duffner, A. M. Lozano, R. R. Tasker, and S. Houle, “Activation of the anterior cingulate cortex by thalamic stimulation in patients with chronic pain,” J. Neurosurg., vol. 92, no. 1, pp. 64–69, Jan. 2000.

[13] B. H. Bewernick, et al., “Nucleus accumbens deep brain stimulation decreases ratings of depression and anxiety in treatment-resistant depression,” Biological Psychiatry, vol. 67, no. 2, pp. 110–116, 2010.

[14] M. L. Kringelbach and K. C. Berridge, “The functional neuroanatomy of pleasure and happiness,” Discov. Med., vol. 9, no. 49, pp. 579–587, Jun. 2010.

[15] ClinicalTrials.gov, “Brain computer interface,” U.S. National Library of Medicine.

[16] J. Rock-Torcivia, “Neuralink reaches 7 patients, enabling paralyzed users to control computers with their minds,” R&D World, Jun. 30, 2025.

[17] Reuters, “Neuralink’s speech restoration device gets FDA’s ‘breakthrough’ tag,” May 1, 2025.

[18] Reuters, “Musk’s Neuralink says second trial implant went well, no thread retraction issue,” Aug. 21, 2024.

[19] Fintool News, “Neuralink to begin high-volume production of brain chips in 2026,” Dec. 31, 2025.

[20] E. Musk et al., “An integrated brain–machine interface platform with thousands of channels,” J. Med. Internet Res., vol. 21, no. 10, 2019.

[21] M. Deutscher, “Neuralink raises $650M at reported $9B pre-money valuation,” SiliconANGLE, Jun. 2, 2025.

[22] Reuters, “Musk’s Neuralink raises $650 million in latest funding round,” Jun. 2, 2025.

[23] Statifacts, “Brain Computer Interface Market – Databook 2025–2034,” Apr. 2025.

[24] R. Zhu, “Paradigm shift in global governance of medical brain-computer interfaces,” Risk Management and Healthcare Policy, 2025.

[25] Council of the European Union, “From vision to reality: Promises and risks of brain-computer interfaces,” Sept. 2024.

OALL Exclusive Data

All analyses marked [OALL Data] in this report are based on the OALL science and technology database, covering exclusive trend analysis across global papers, patents, funding events, and company data. Data sources include but are not limited to: cross-analysis of PubMed, Web of Science, Scopus, USPTO, WIPO, Crunchbase, and PitchBook, as well as OALL's proprietary AI semantic analysis engine.

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